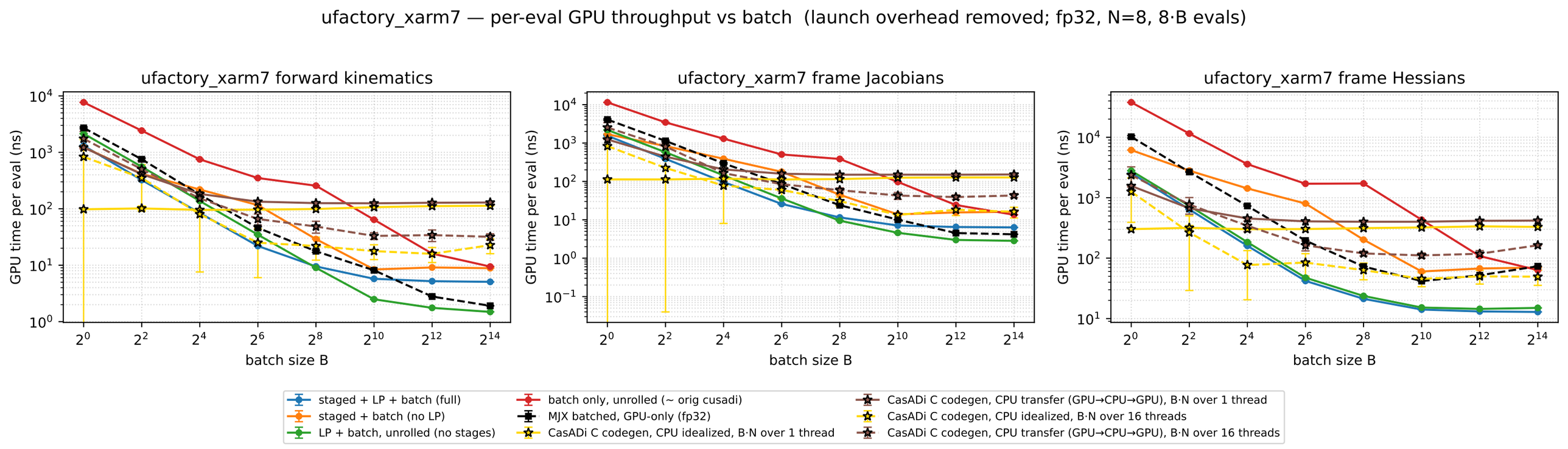

ufactory_xarm7 — per-eval GPU throughput vs batch (launch overhead removed; fp32, N=8, 8·B evals)
ufactory_xarm7 forward kinematics
ufactory_xarm7 frame Jacobians
ufactory_xarm7 frame Hessians
GPU time per eval (ns)
batch size B
staged + LP + batch (full)
staged + batch (no LP)
LP + batch, unrolled (no stages)
batch only, unrolled (~ orig cusadi)
MJX batched, GPU-only (fp32)
CasADi C codegen, CPU idealized, B·N over 1 thread
CasADi C codegen, CPU transfer (GPU→CPU→GPU), B·N over 1 thread
CasADi C codegen, CPU idealized, B·N over 16 threads
CasADi C codegen, CPU transfer (GPU→CPU→GPU), B·N over 16 threads